

2-Phase Interleaved Sync-Buck Converter

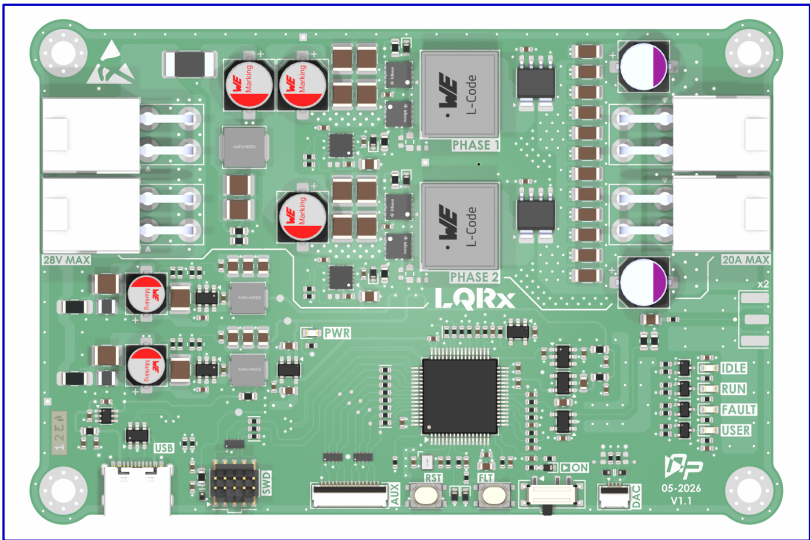
Digitally Controlled Power Conversion with STM32G474

Revision 1.1

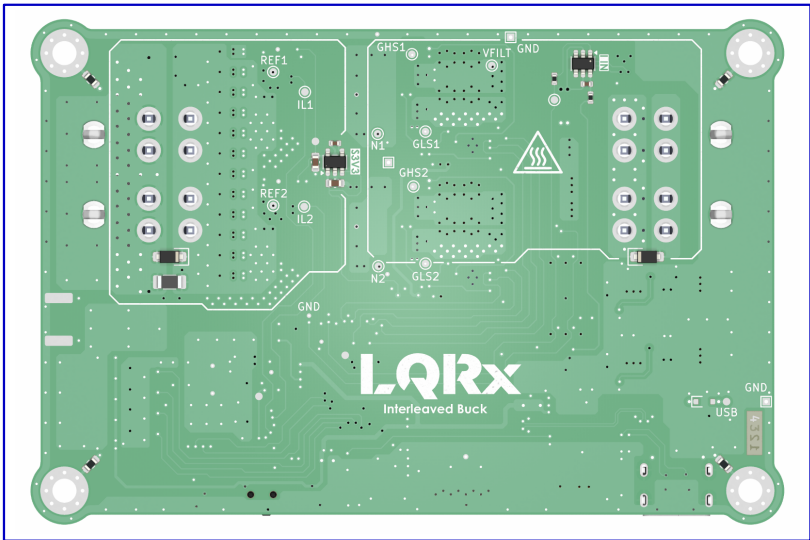
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- B Block Diagram
- C Top Hierarchical View
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TOP VIEW



BOTTOM VIEW



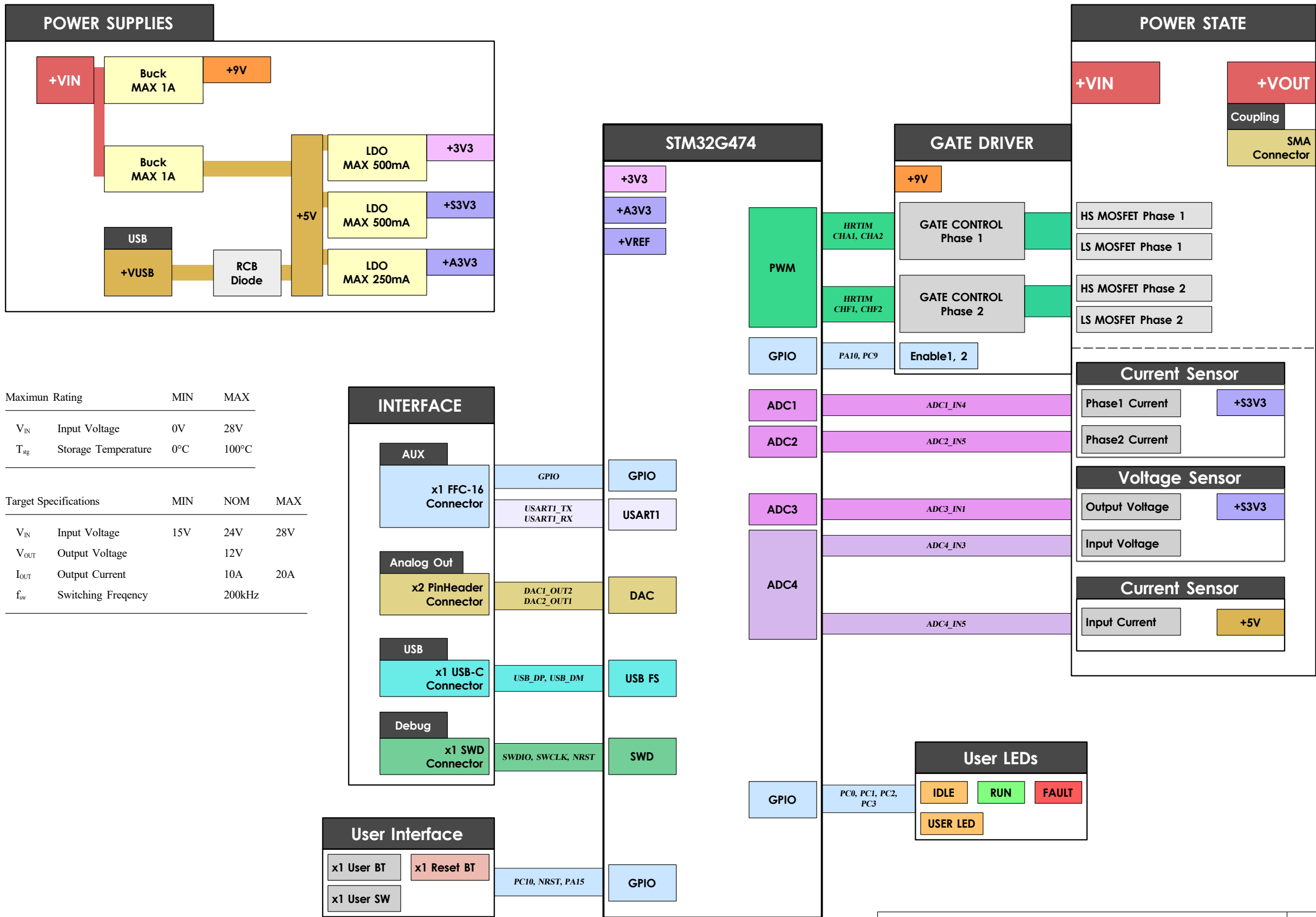
Legend

DESIGN NOTE: Text for information design.	DEBUG NOTE: Text for Debug notes.	DESIGN NOTE: Text for design guideline	DESIGN NOTE: Text for critical design	LAYOUT NOTE: Text for layout guideline	LAYOUT NOTE: Text for Critical layout	General Note Warning Note Text to be added to silkscreen.
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LQRx

Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Project Overview Cover Page	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.1 Date: 28/06/2026
	Sheet: /	Size: A3	Sheet: A of 9
KiCad E.D.A. 10.0.6		File: 2Phase_SynchBuck.kicad_sch	

[B] Block Diagram

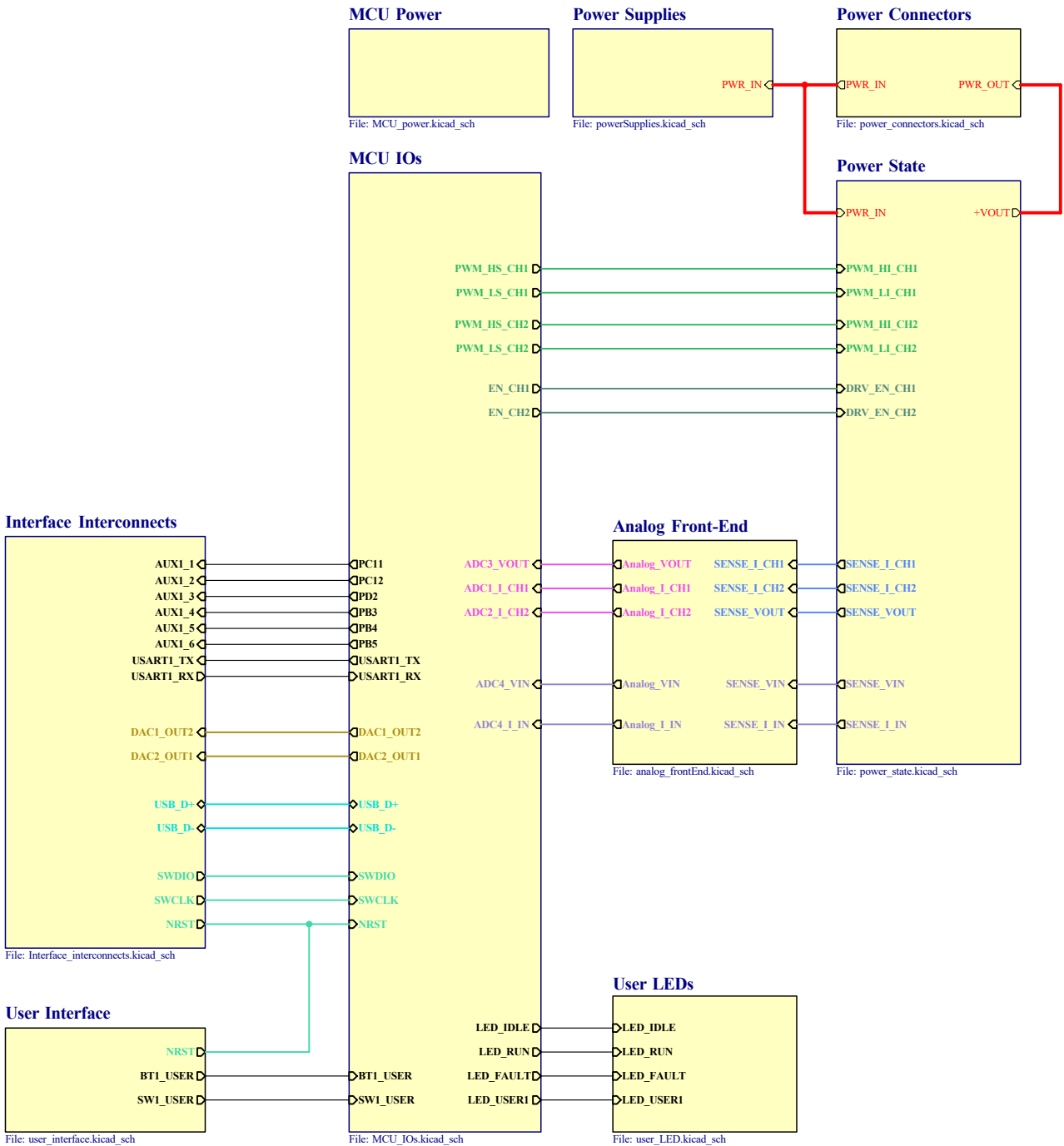


Maximun Rating		MIN	MAX
V _{IN}	Input Voltage	0V	28V
T _{stg}	Storage Temperature	0°C	100°C

Target Specifications		MIN	NOM	MAX
V _{IN}	Input Voltage	15V	24V	28V
V _{OUT}	Output Voltage		12V	
I _{OUT}	Output Current		10A	20A
f _{sw}	Switching Frequency		200kHz	

Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Block Diagram (Top Level View)	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.1 Date: 28/06/2026
	Sheet: /Block Diagram/		Size A3
	File: Block_diagram.kicad_sch		Sheet: B of 9
KiCad E.D.A. 10.0.6			

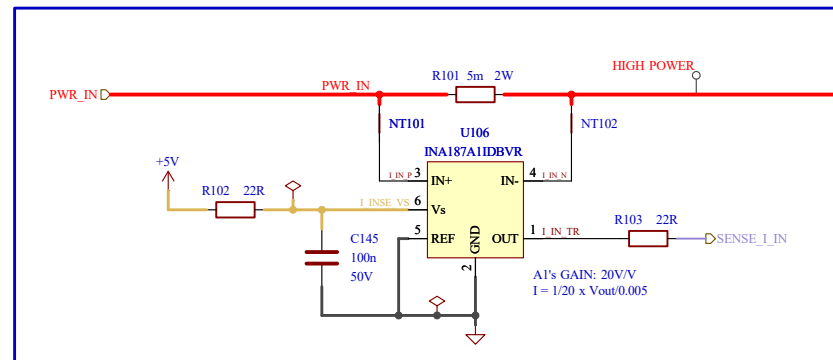
[C] Top Hierarchical View



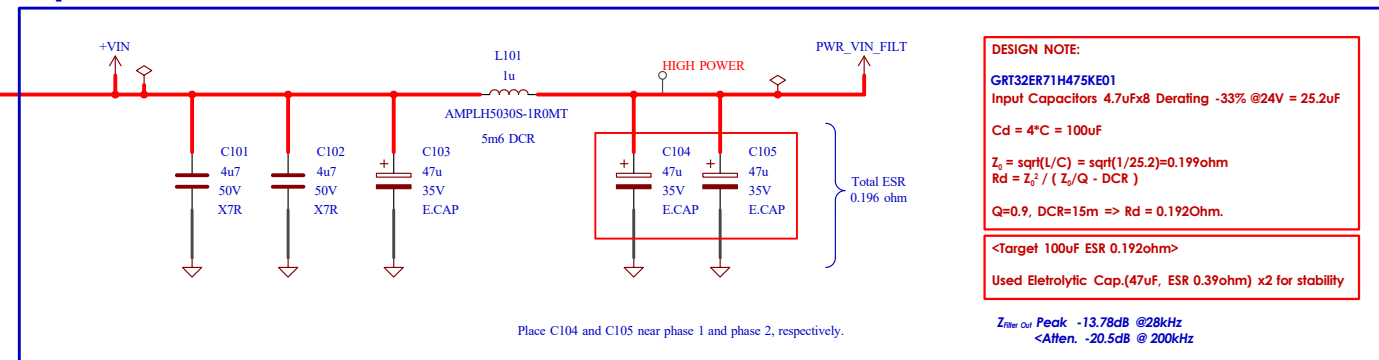
Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Top Hierarchical View (Top Level View)	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.1 Date: 28/06/2026
	Sheet: /Top/	Size: A3	Sheet: C of 9
KiCad E.D.A. 10.0.6		File: Top_hierarchical.kicad_sch	

[1] Power State

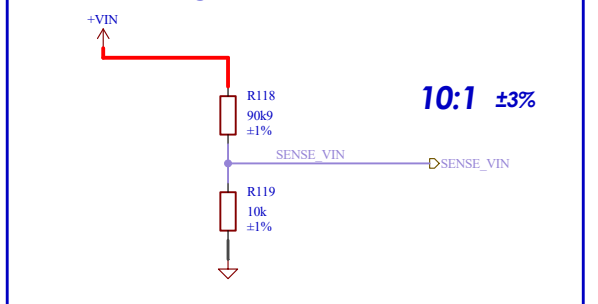
I_{IN} Sense



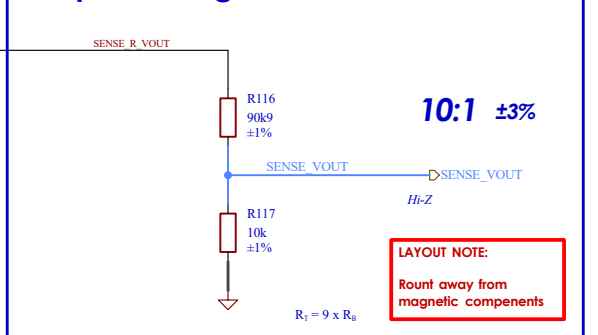
Input Filter



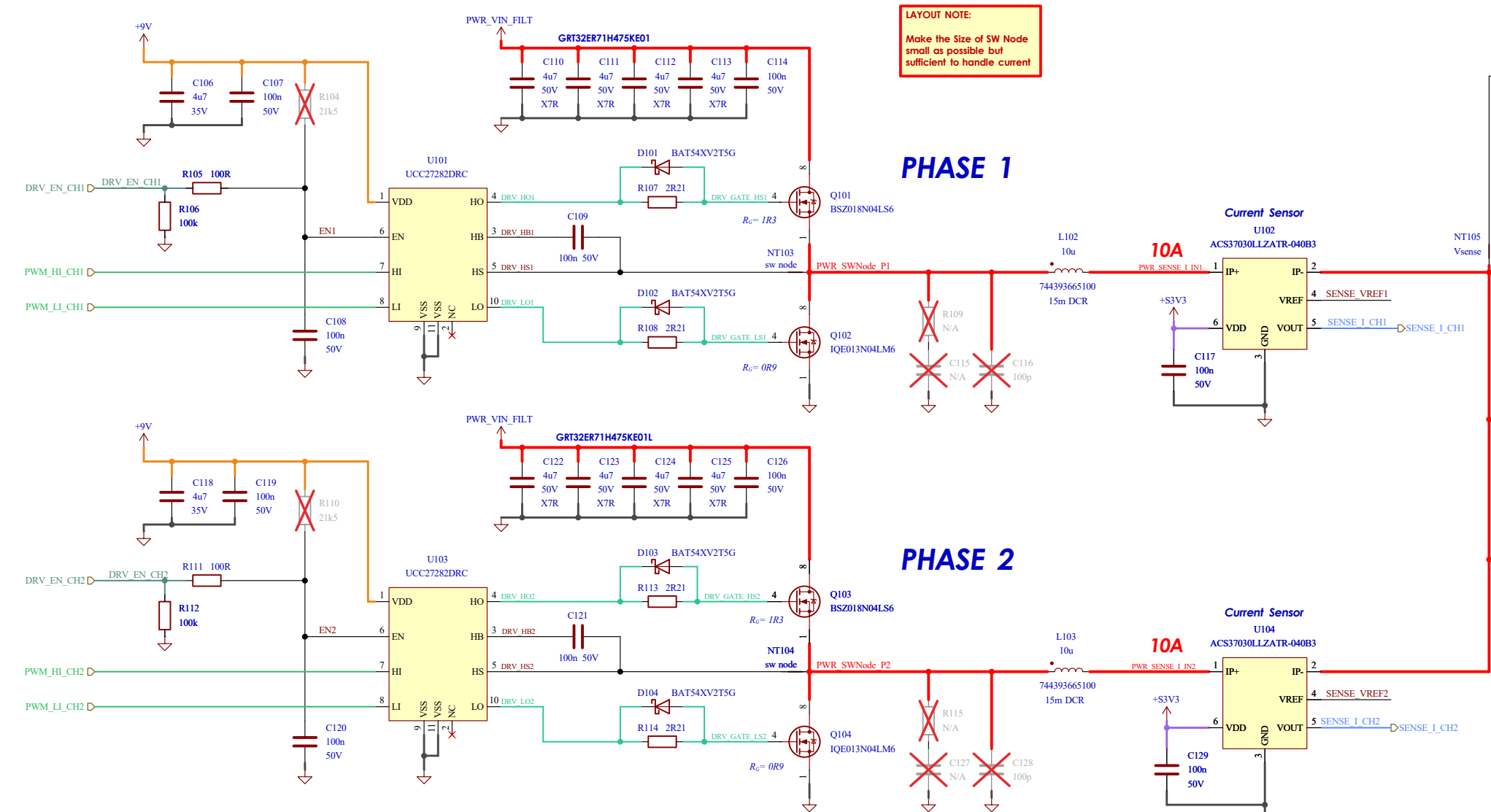
Input Voltage Sense



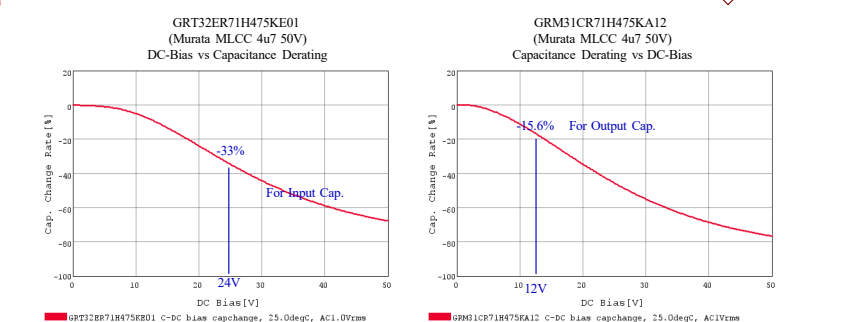
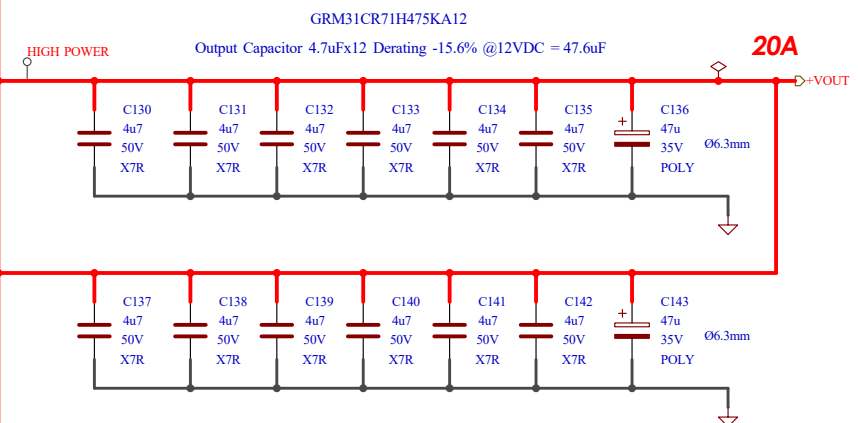
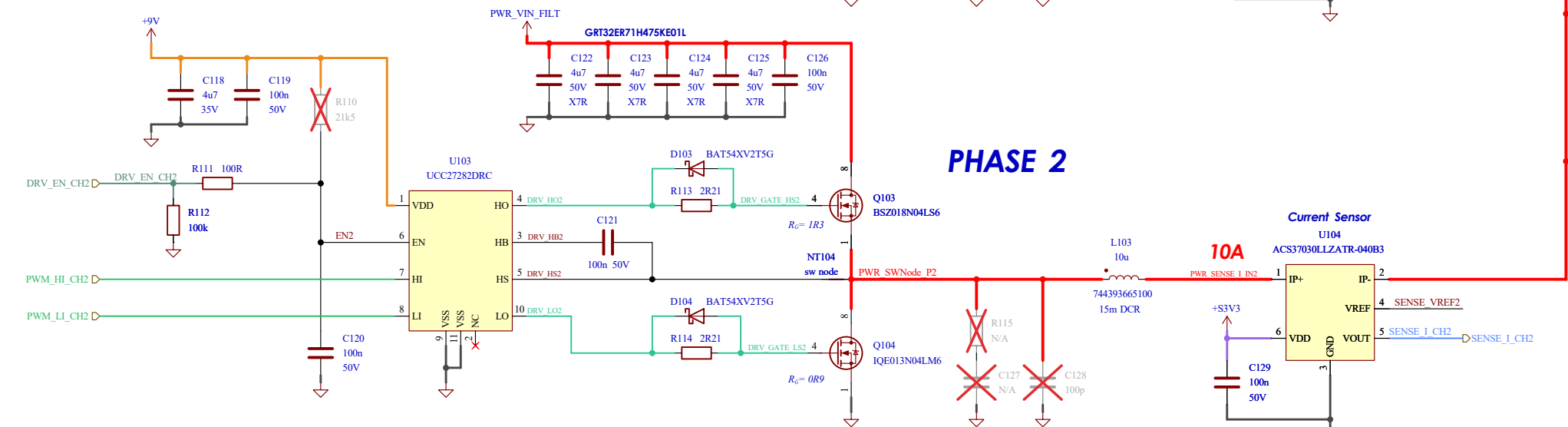
Output Voltage Sense



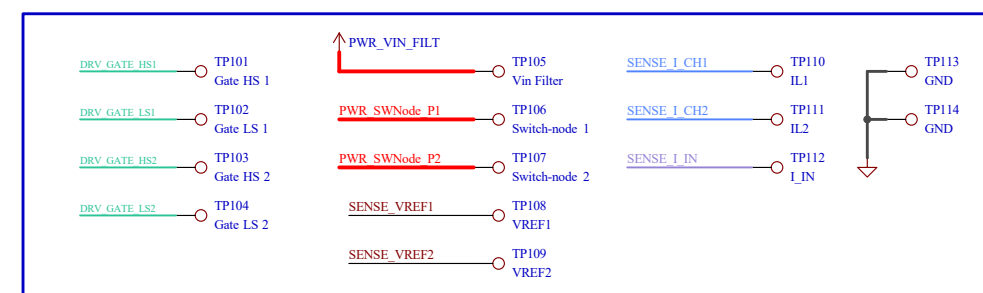
PHASE 1



PHASE 2



Test Point

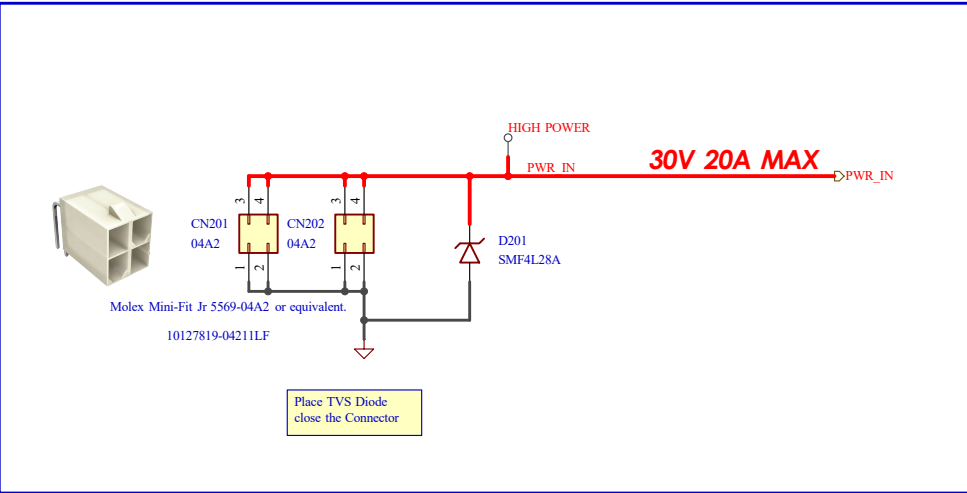


DEBUG NOTE:
If Switch-node are ringing
C116 and C128 are capacitor to find R C values in the snubber circuit.
 $C_{sn} = C_{gr} [f_s^2 / (f_s^2 - f_s^2)]$
 $L_{sn} = 1 / [2\pi f_s C_{sn}]$
 $C_{snb} = (4 \cdot 8) C_{gr}$
 $R_{snb} = 1/Q [\sqrt{L_{sn} / C_{snb}}]$

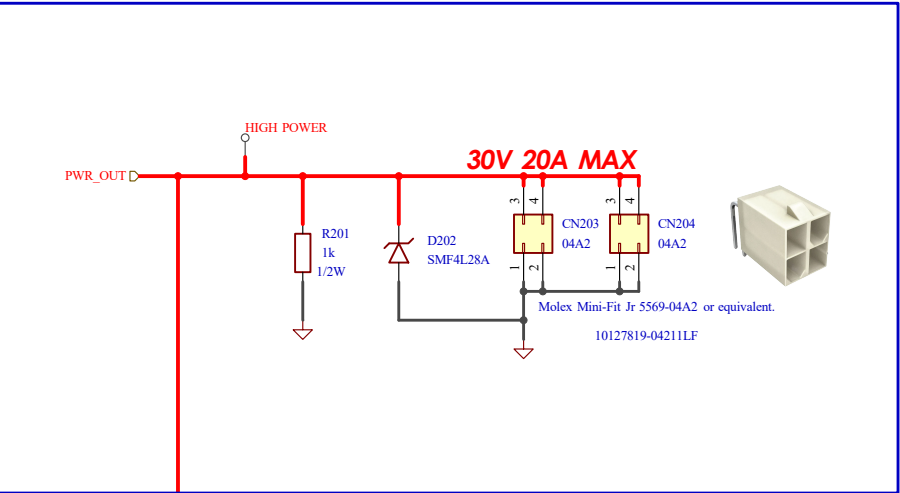
Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Power State 2-Phase Buck & Gate Driver	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.1
Sheet: /Top/Power State/ File: power_state.kicad_sch		Size: A3	Date: 28/06/2026
KiCad E.D.A. 10.0.6		1 of 9	

[2] Power Connectors

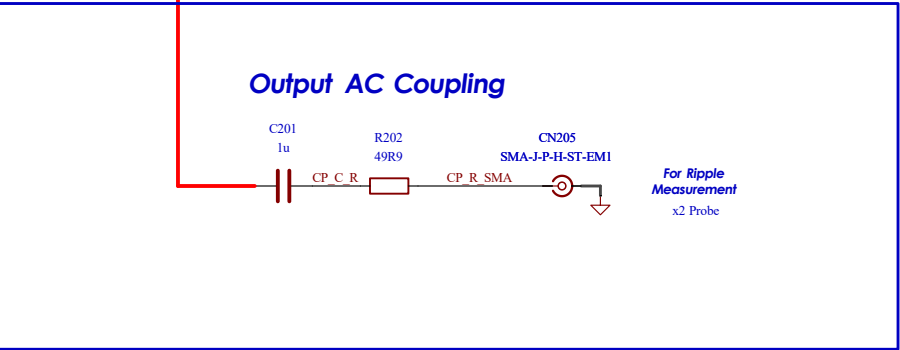
Input Connector



Output Connector



Output AC Coupling

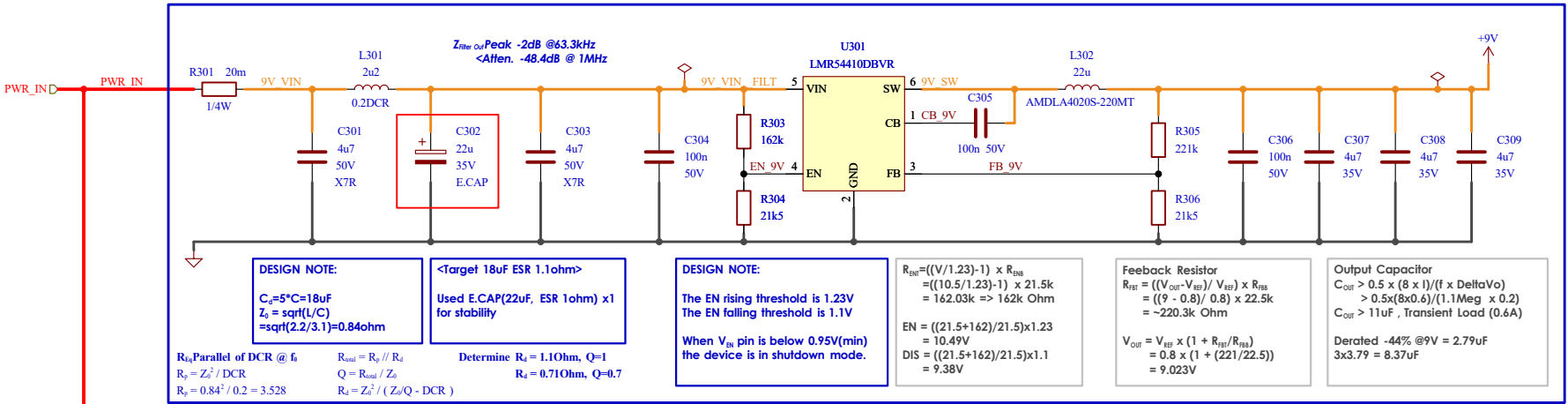


Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Power Connectors	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.1 Date: 28/06/2026
	Sheet: /Top/Power Connectors/		Size A3
KiCad E.D.A. 10.0.6	File: power_connectors.kicad_sch		Sheet: 2 of 9

[3] Power Supplies

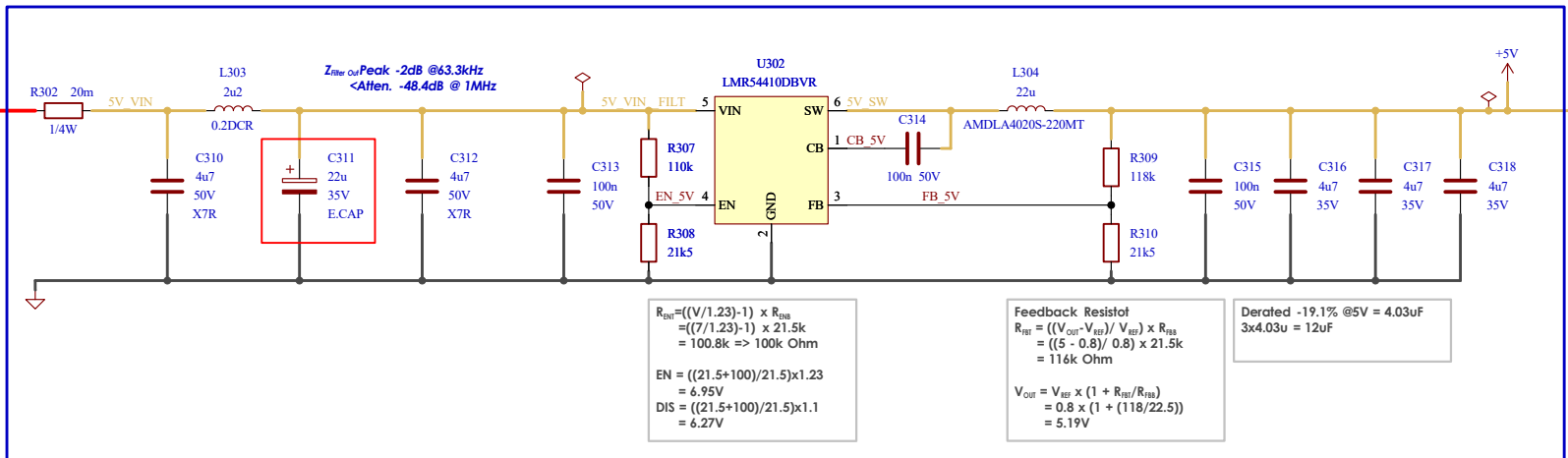
Buck Converter 9V 1A (1.1MHz)

*Enable when V_{IN} is above 11V

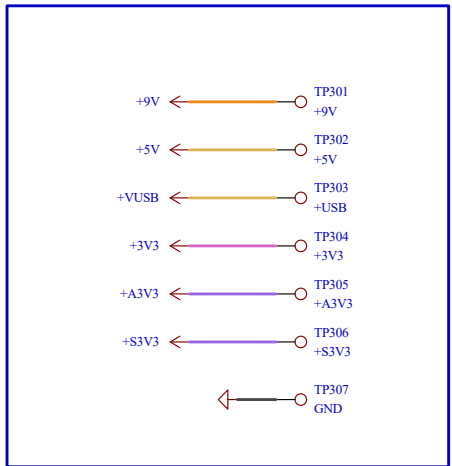


Buck Converter 5V 1A (1.1MHz)

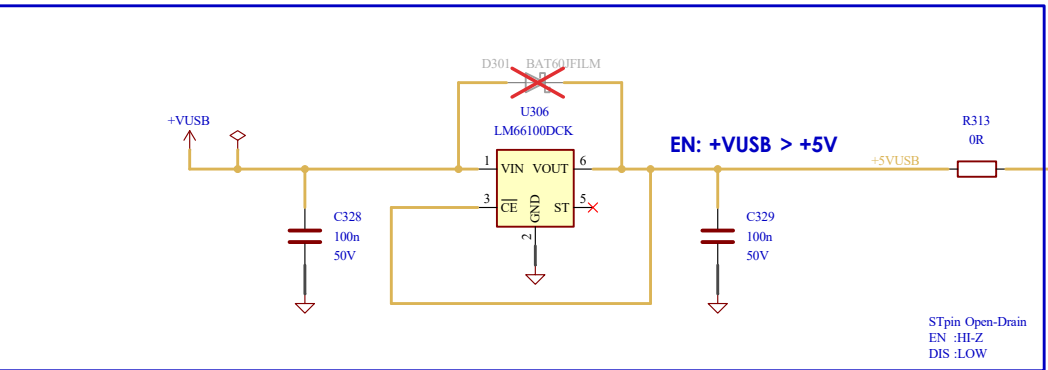
*Enable when V_{IN} is above 7V



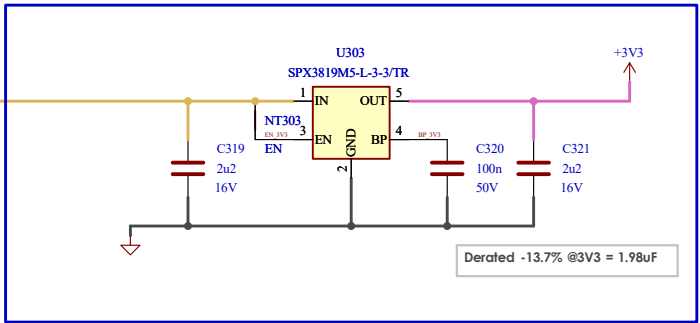
Test Point



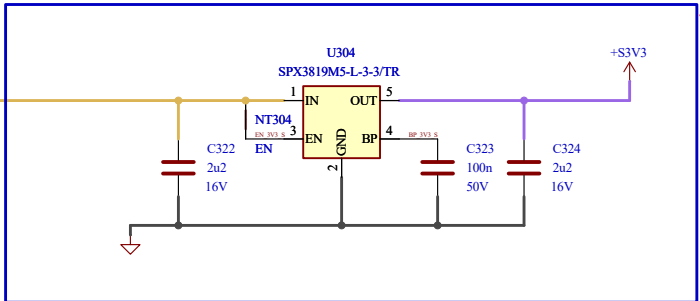
Reverse Current Blocking



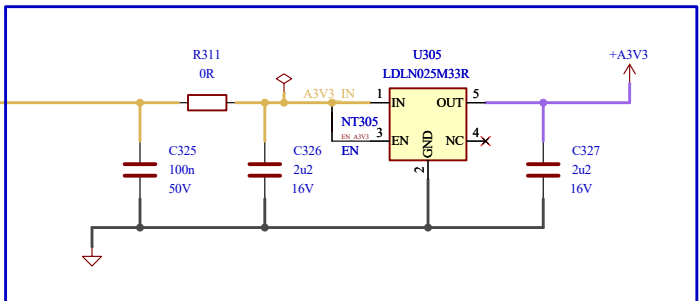
LDO 3V3 500mA (Digital path)



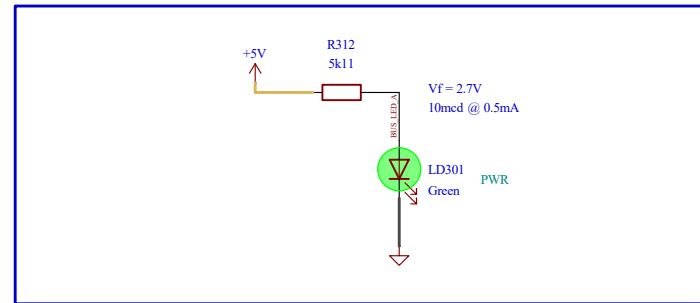
LDO 3V3 500mA (Sensor path)



LDO 3V3 250mA (Analog path)

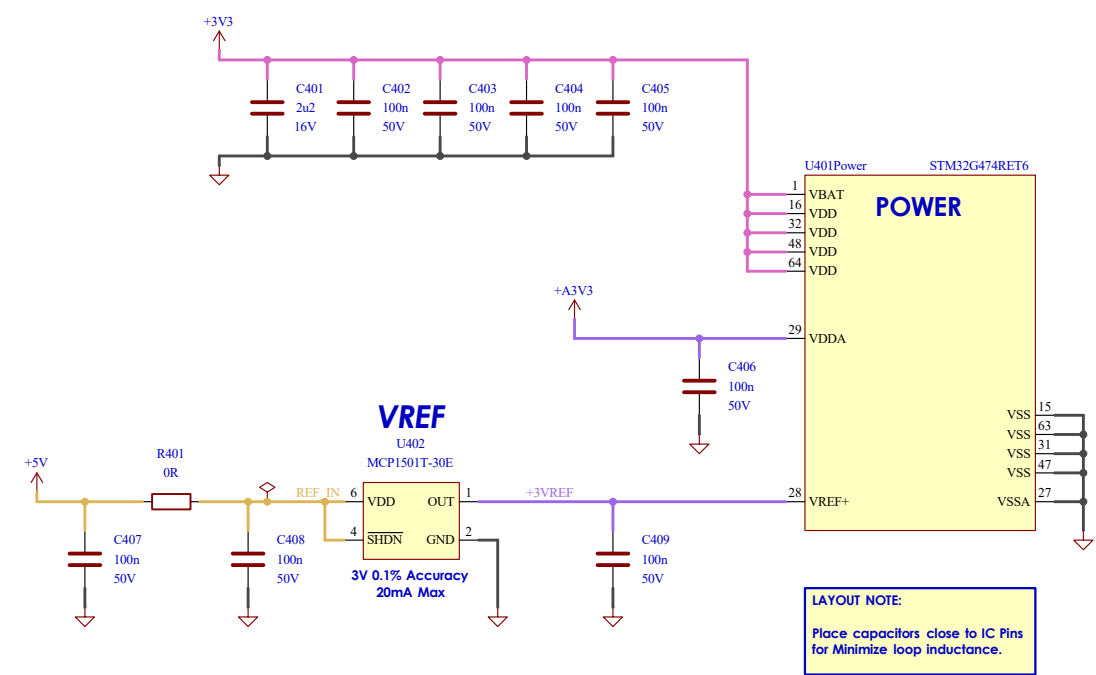


LED Indicators



Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Power Supplies Auxiliary 9V 5V 3V3	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.1 Date: 28/06/2026
Sheet: /Top/Power Supplies/ File: powerSupplies.kicad_sch		Size: A3	Sheet: 3 of 9
KiCad E.D.A. 10.0.6			

[4] Microcontroller Power



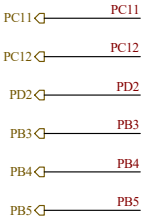
Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Microcontroller Power STM32G474Rx	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.1 Date: 28/06/2026
	Sheet: /Top/MCU Power/		Size A3
KiCad E.D.A. 10.0.6	File: MCU_power.kicad_sch		Sheet: 4 of 9

[5] Microcontroller IOs

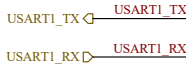
Interconnects

Auxiliary Port 1

GPIO



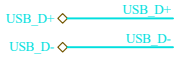
USART



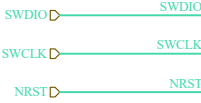
Analog Output (DAC)



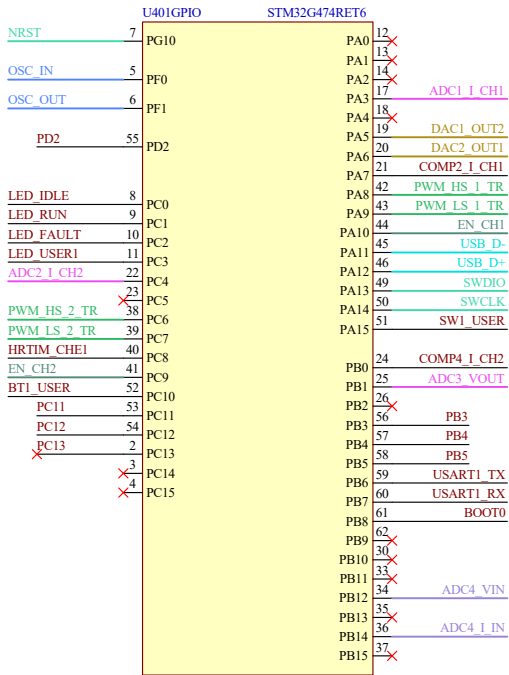
USB2.0 FS



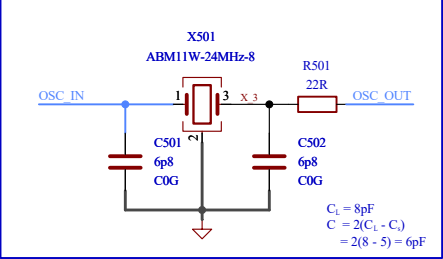
Serial Wire Debug



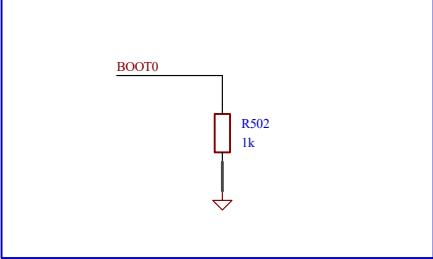
STM32G474 I/O



HSE Clock (24MHz)



BOOT0



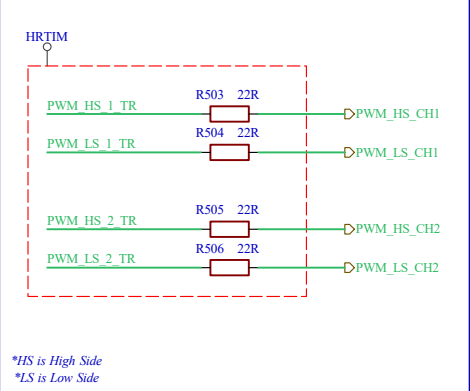
User Interface



Test Point



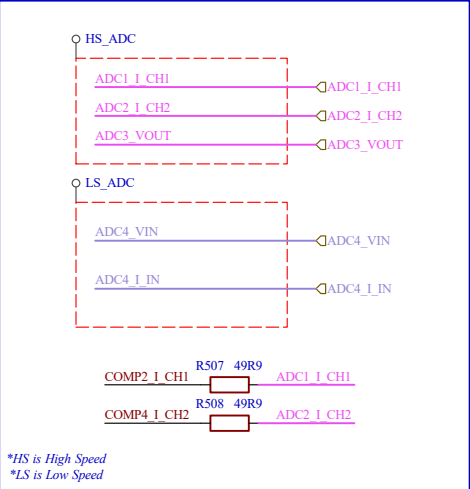
High Resolution Timer



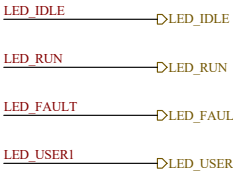
Enable Gate Driver



ADC



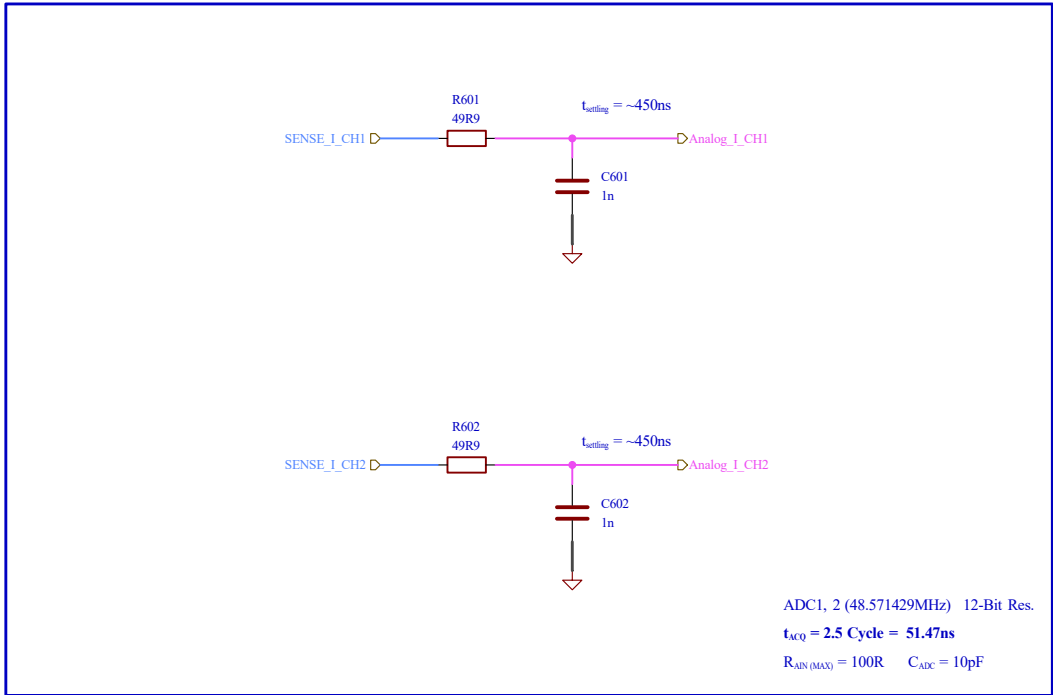
LEDs



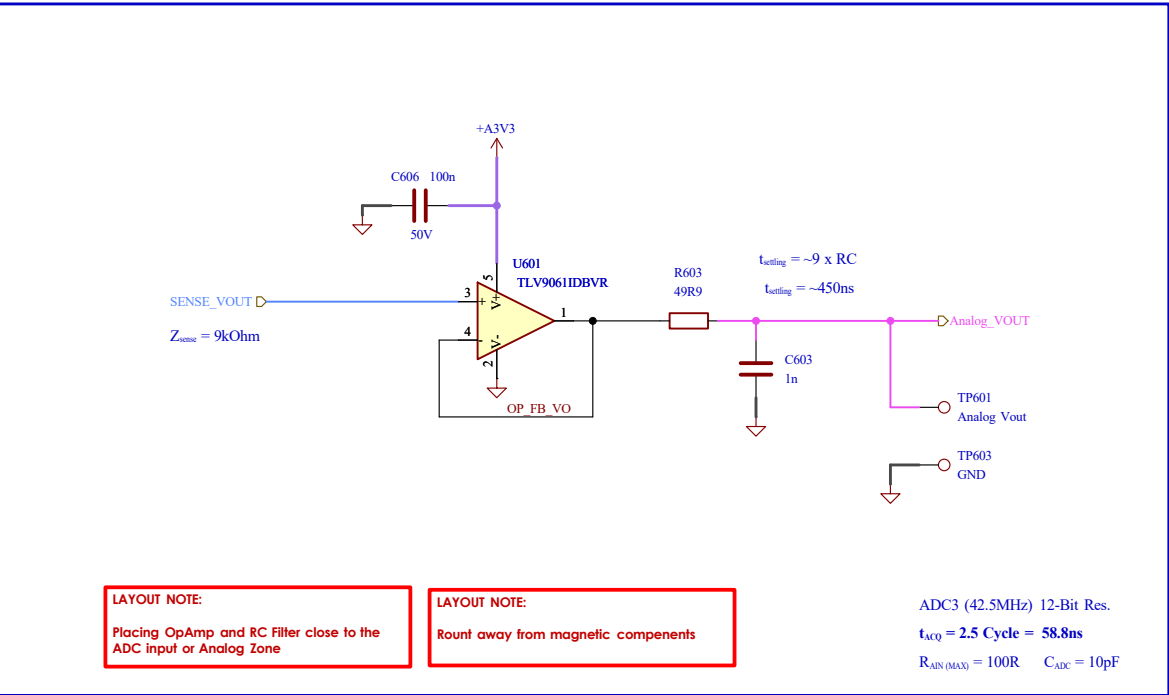
Project: 2-Phase Interleaved Sync-Buck Converter				LQRx		
Title: Microcontroller IOs STM32G474Rx	Drawing by: NATCHPAI		Date: 08/04/2026		Variant:	
	Reviewed by:		Date:		Rev: 1.1	Date: 28/06/2026
	Sheet: /Top/MCU IOs/ File: MCU_IOs.kicad_sch				Size A3	Sheet: 5 of 9
	KiCad E.D.A. 10.0.6					

[6] Analog Front-End

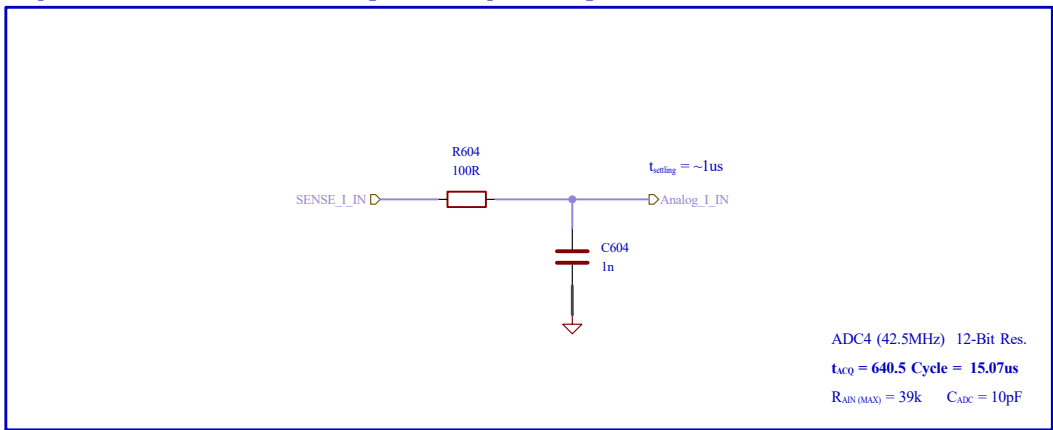
Current Path (High Speed)



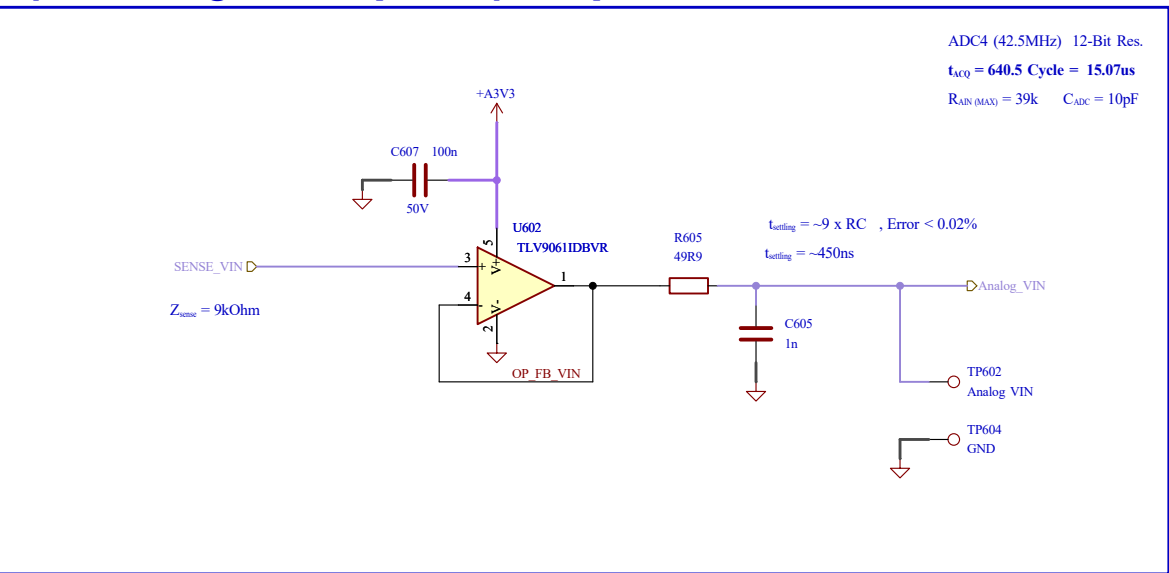
Output Voltage Path (High Speed)



Input Current Path (Low Speed)



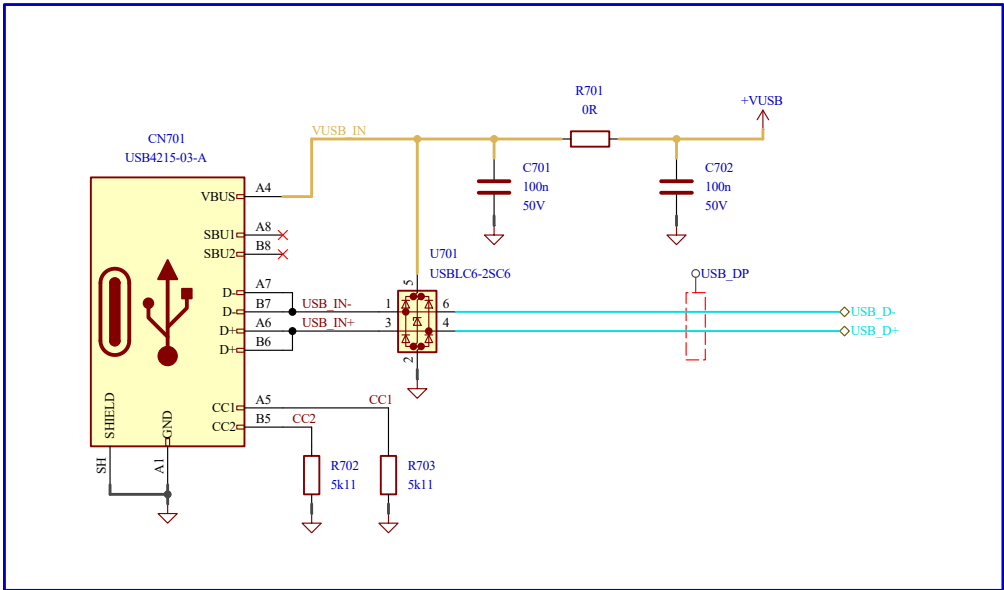
Input Voltage Path (Low Speed)



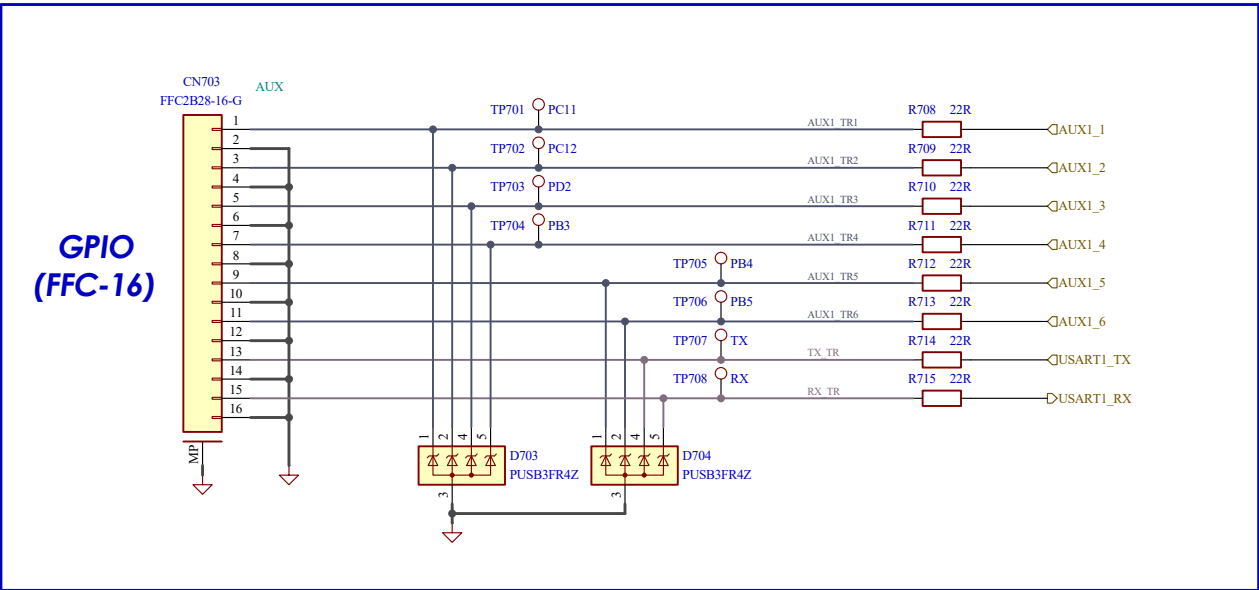
Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Analog Front-End	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.1 Date: 28/06/2026
	Sheet: /Top/Analog Front-End/ File: analog_frontEnd.kicad_sch		Size A3 Sheet: 6 of 9
KiCad E.D.A. 10.0.6			

[7] Interface Interconnects

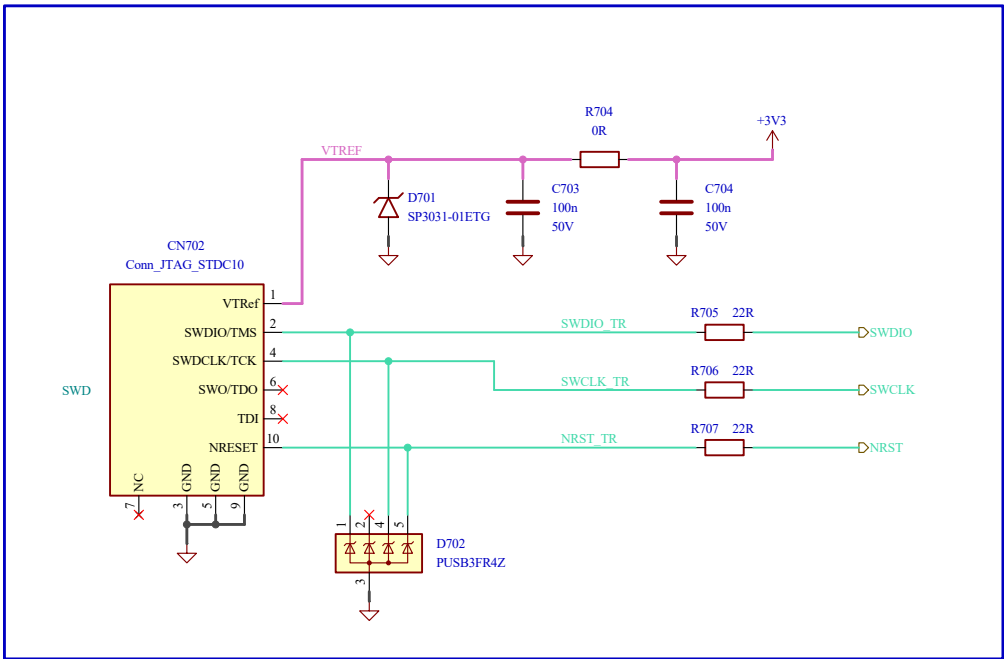
USB-C Connector



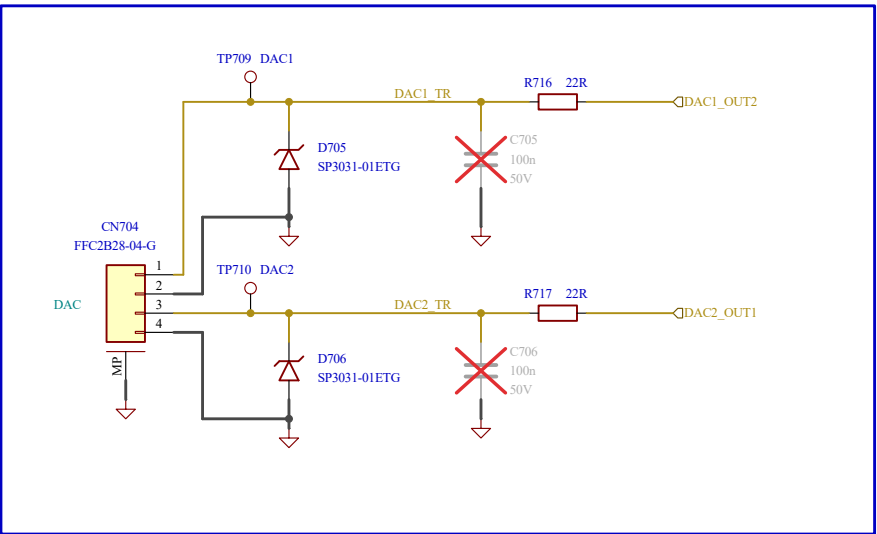
AUX Connector



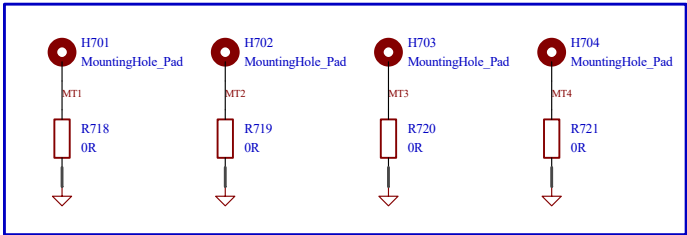
SWD Connector



Analog Output



Mounting Hole

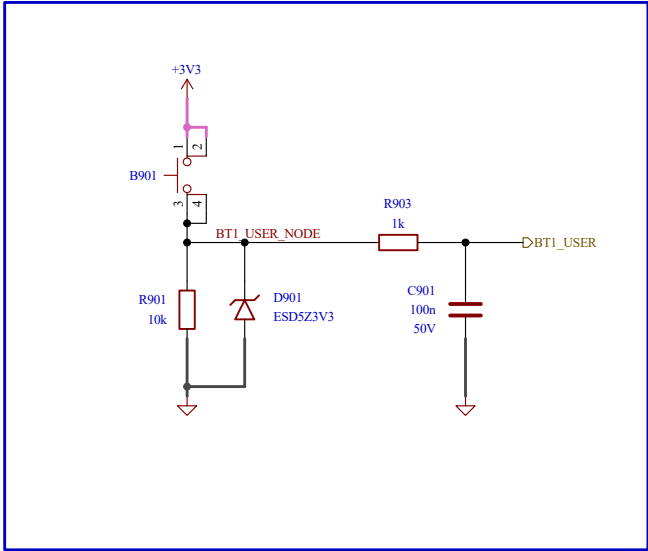


Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Interface Interconnects	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.1 Date: 28/06/2026
	Sheet: /Top/Interface Interconnects/ File: Interface_interconnects.kicad_sch		Size: A3 Sheet: 7 of 9
KiCad E.D.A. 10.0.6			

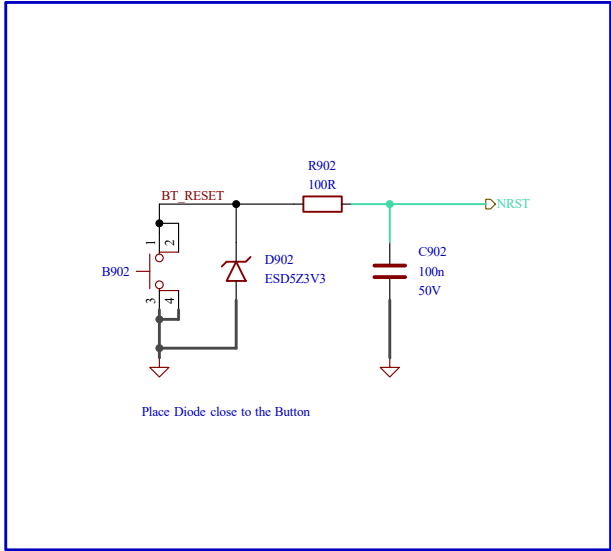
Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: User LEDs Status LEDs	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.1 Date: 28/06/2026
	Sheet: /Top/User LEDs/		Size A3
KiCad E.D.A. 10.0.6	File: user_LED.kicad_sch		

[9] User Interface

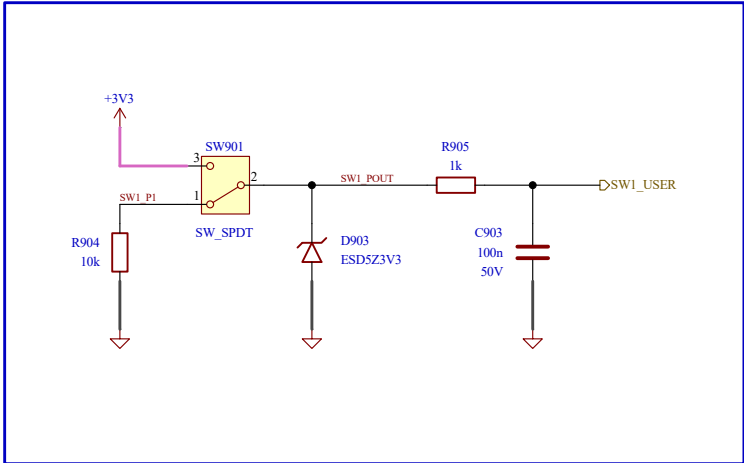
User Button



Reset Button

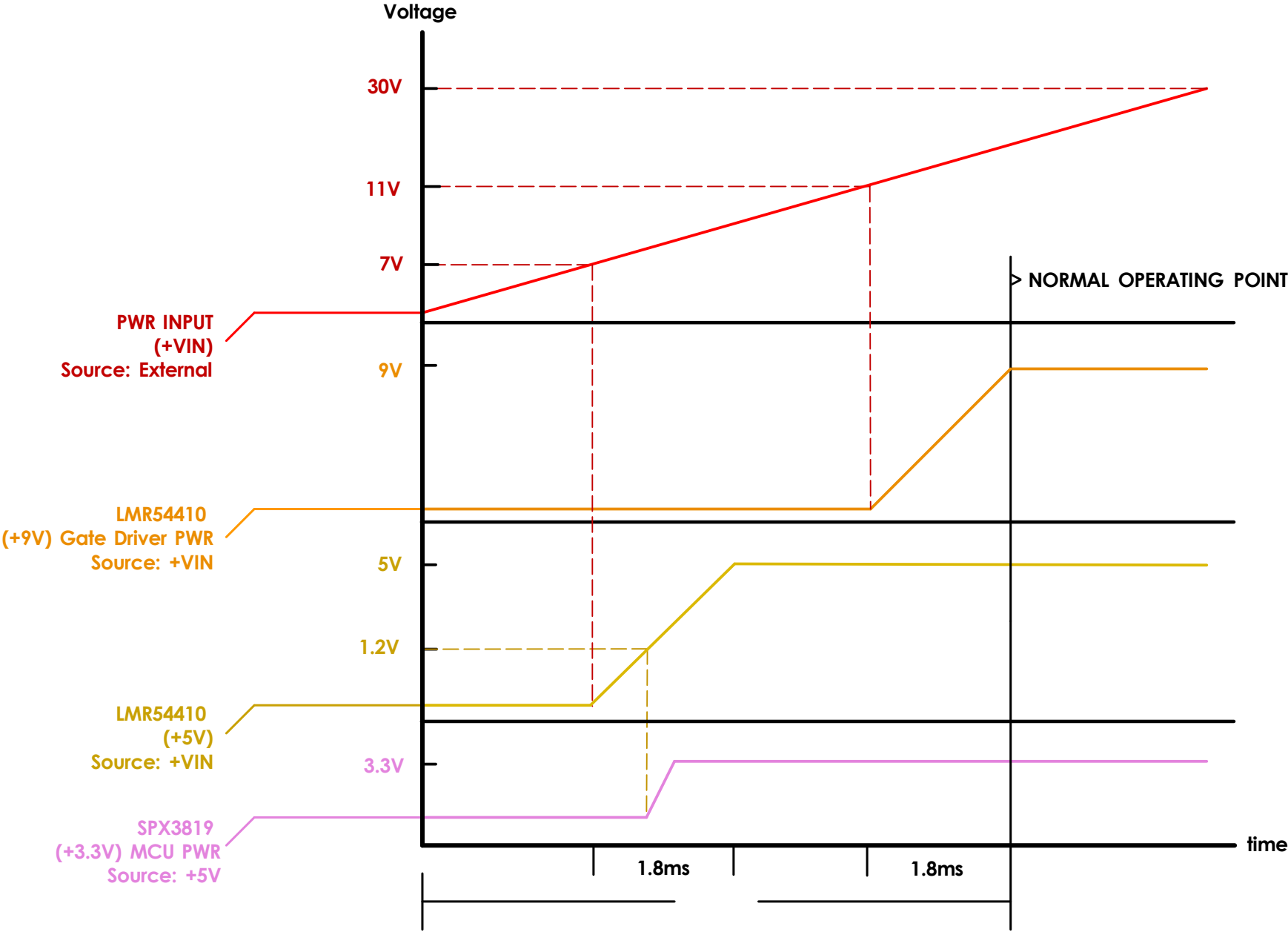


User Switch



Project: 2-Phase Interleaved Sync-Buck Converter		LQRx		
Title: User Interface Button & Switch	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:	
	Reviewed by:	Date:	Rev: 1.1	Date: 28/06/2026
	Sheet: /Top/User Interface/		Size A3	Sheet: 9 of 9
	KiCad E.D.A. 10.0.6		File: user_interface.kicad_sch	

[D] Power Supply Sequencing



Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Power Supply Sequencing	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.1 Date: 28/06/2026
	Sheet: /Power Supply Sequencing/		Size A3 Sheet: D of 9
KiCad E.D.A. 10.0.6		File: pwr_sequencing.kicad_sch	

[E] Revision History

Revision 1.0

- (18/05/2026) Initialize the project
- (03/06/2026) PC Board is Released

Revision 1.1

- (28/06/2026)
- Remove the MOSFETs that Enable the ICs Gate Driver and connect directly. (p.1)
 - Change ADC input Capacitor 47pF to 1nF. (p.6)
 - Change OP-AMP U601 U602 to correctly part number (p.6)

Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Revision History	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.1 Date: 28/06/2026
	Sheet: /Revision History/		Size A3
KiCad E.D.A. 10.0.6	File: RevHistory.kicad_sch		Sheet: E of 9